

SENG 401: Software Architecture

Project Proposal

Project Title: Climate Crusade

United Nations Sustainable Development Goal: SDG #13 – Climate Action

Team Members & Roles:

- Ebitimi Dambo - Game Designer, Art Director, Database Engineer, Project Manager
- Arol Nokam Wafo - Game Designer, Test Engineer
- Aadil Bashir - Game Designer, Database Engineer, Sound Designer
- Stephan Agwai - Game Designer, Test Engineer
- Gauthier Appaix - Game Designer, Sound Designer
- Emmanuel Idoro - Game Designer, Quality Engineer, Test Engineer

Project Idea

This game follows an extraterrestrial species member named Alvin, who comes from a galactic colony dedicated to helping planets on the brink of collapse due to global warming. The colony previously attempted to save a planet called Pluto, but the mission failed, causing the planet to explode. The resulting impact sent Alvin drifting through space until he crash-lands on Earth. Upon arrival, Alvin discovers that Earth's life force is already at 50% and rapidly declining. He decides to intervene and attempt to save the planet before it reaches the same outcome as Pluto. The game focuses on the 13th United Nations Sustainable Development Goal, Climate Action. Players follow Alvin as he travels to different countries around the world that are being affected by climate change, learning about the different ways each country is impacted. Climate change appears in many forms, including acid rain, polluted waters, dried-out deserts, and falling space debris. The game features two playable levels, with possible surprise levels included if completed by the submission deadline. Each level uses unique gameplay mechanics, such as platforming, dig-site exploration, and shooter-style interactions, to explore these climate issues. Players help Alvin reduce the damage caused by climate change while learning practical do's and don'ts related to each problem. The intended audience is individuals who are open to change. The game aims to encourage these players to take greater responsibility for protecting the planet.

Technological Stack

The game will be developed using Phaser.js, a JavaScript-based 2D game framework that runs in the browser. Phaser.js will serve as the front end and will communicate with the back end by sending HTTP requests to retrieve and store data in an SQL database. The back end will be built using Node.js, which handles request validation and executes SQL queries to manage game data. Since Phaser.js is an HTML5 framework that uses JavaScript along with HTML and CSS for UI integration, the game will be deployed as a web-based application. This allows it to run on any modern web browser, ensuring cross-platform compatibility across desktop and mobile devices.

Team Roles

The team consists of five members with clearly defined responsibilities. Ebitimi serves as the Project Manager and is responsible for overseeing the project timeline and resources. He is also in charge of art assets, database creation, and designing one of the game levels. Arol is responsible for designing a level and conducting routine and extensive testing throughout different stages of development. Aadil is responsible for designing a level, assisting with database creation, and handling sound design. Stephan is responsible for designing a level and regularly testing the game alongside Arol. Gauthier is responsible for designing a level and serves as the primary sound designer. He also supports testing efforts when needed. Emmanuel assists Stephan with level design and serves as a quality engineer, ensuring the game features are up to standard.